Towards Cocoon: A Simulation Scaffold for Citizen-Co-Created Low-Height Distributed MIMO Access Networks

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***Abstract*—This paper studies low-height distributed-multiple-input multiple-output (MIMO) access networks with repositionable access points in an outdoor urban setting. Using the `COCOON-Sionna` framework, the workflow generates wall-mounted candidate access points (APs), simulates pedestrian and bicycle mobility, computes path-based channel state information (CSI), and compares a central rooftop proxy, a fixed distributed deployment, and a relocation-capable distributed deployment over a common scenario. The article formalizes the signal-to-interference-plus-noise ratio (SINR)- and ergodic sum rate (ESR)-based evaluation model, the history-aware relocation heuristic, and the postprocessing workflow that produces geometry, performance, and relocation-cost figures from reproducible simulation outputs.**

***Index Terms*—cell-free systems, distributed MIMO, low-height access points, access-point placement, urban mobility, Sionna RT**

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Fig. 1: Conceptual illustration of the Cocoon low-height access-network vision.

I. Introduction

The Cocoon project targets a wireless access paradigm in which the network adapts to the fluidity of the city rather than assuming a static infrastructure and a stationary demand profile. The motivating hypothesis from the project proposal is that a practical cell-free (CF) network becomes more feasible when low-height infrastructure can be co-created, repositioned, and activated only when the measured channel conditions justify it. In that view, the citizens are not only traffic sources but also anchors for sensing, mounting, and reconfiguring the surrounding network.

This article narrows that broader vision to a simulation-based study of low-height infrastructure adaptation. The implemented framework covers wall-mounted candidate access points (APs), outdoor user mobility, path-based channel state information (CSI) extraction, and repeated placement decisions over relocation windows in an urban scene. This scope enables a reproducible comparison of geometry, radio-performance, and relocation-cost metrics for alternative infrastructure strategies.

The main contributions of the present study are:

- a concise technical framing of Cocoon as a low-height, movable-AP, citizen-co-created access-network concept,
- a reproducible description of the Rabot outdoor simulation pipeline implemented in `COCOON-Sionna`, and
- a common analysis workflow for geometry, signal-to-interference-plus-noise ratio (SINR), ergodic sum rate (ESR), and relocation-cost results derived from exported simulation artifacts.

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II. Scope Within Cocoon

The full Cocoon proposal spans citizen interaction with the wireless network, channel assessment, architecture design, and proof-of-concept validation. The present article addresses the simulation-facing part of that program, with emphasis on dynamic channel-aware adaptation, low-height AP/contact service point (CSP) placement, and the algorithmic consequences of allowing physical repositioning.

The main modeling simplification is that co-creation is represented here through an abstract movable infrastructure layer rather than through a user study or a deployment trial. Candidate mounting locations stand in for positions that could later be offered to citizens, and the placement scheduler stands in for the future tool chain that would notify participants where the next best CSP constellation should be installed.

Accordingly, the scope of the present article is:

- the paper validates a simulation methodology and a comparison workflow, not the social feasibility of citizen participation,
- the paper studies a Rabot outdoor scenario, not yet a multi-site or multi-season campaign,

- the paper compares placement strategies over a common CSI-derived objective, not yet a full end-to-end network stack.

III. System Model

We consider a downlink distributed MIMO (D-MIMO) system with active APs $\mathcal{A} = \{1, \ldots, M\}$ and scheduled user equipments (UEs) $\mathcal{U} = \{1, \ldots, K\}$. Let N_m denote the number of transmit ports at AP m and $N_{r,k}$ the number of receive ports at UE k. The per-AP channel block is

$$\mathbf{H}_{k,m} \in \mathbb{C}^{N_{r,k} \times N_m}, \tag{1}$$

and the full distributed channel observed by UE k is

$$\mathbf{H}_k = [\mathbf{H}_{k,1}\ \mathbf{H}_{k,2}\ \cdots\ \mathbf{H}_{k,M}]. \tag{2}$$

Given a receive combiner $\mathbf{u}_k$, the effective channel becomes

$$\mathbf{h}_k^{\mathsf{H}} = \mathbf{u}_k^{\mathsf{H}} \mathbf{H}_k. \tag{3}$$

Stacking all users yields the effective distributed channel matrix $\mathbf{H}_{\text{eff}}$. The current pipeline applies a least-squares pseudo-inverse zero-forcing (ZF) precoder on $\mathbf{H}_{\text{eff}}$, followed by per-stream power allocation under a total AP power budget.

For UE k, the instantaneous downlink SINR is evaluated as

$$\text{SINR}_k^{\text{DL}} = \frac{p_k \left|\mathbf{h}_k^{\mathsf{H}} \mathbf{f}_k\right|^2}{\sum_{i \in \mathcal{U},\, i \neq k} p_i \left|\mathbf{h}_k^{\mathsf{H}} \mathbf{f}_i\right|^2 + \sigma^2}, \tag{4}$$

where $\mathbf{f}_i$ is the precoding vector for stream i, p_i its allocated power, and σ^2 the thermal-noise term. The implementation keeps the interference term explicitly and exports desired-signal, interference, and noise power terms for each evaluated user snapshot.

The ESR used later in the results section is derived from the same per-snapshot SINR values. For snapshot n with scheduled-user set $\mathcal{U}_n$, the postprocessing layer computes

$$\text{ESR}_n = \sum_{k \in \mathcal{U}_n} \log_2\left(1 + \text{SINR}_{k,n}^{\text{DL}}\right). \tag{5}$$

These per-snapshot SINR values form the basis of the trajectory-level SINR and ESR statistics reported later in the article.

A. Placement Strategies

The placement layer compares three strategies under a common AP budget:

- `central_massive_mimo`: one rooftop proxy AP is placed on the building whose rooftop representative point is nearest the area center; the proxy is oriented toward the area center. This is a normalized central reference, not a full single co-located-array placement model.
- `distributed_fixed`: a distributed set of wall-mounted candidate positions is sampled once and kept fixed over the full trajectory.
- `distributed_movable`: the same number of distributed APs is reselected at each relocation window from the wall-mounted candidate pool.

Let $\mathcal{C}$ denote the wall-mounted candidate pool and let $\mathcal{S}_w \subseteq \mathcal{C}$ denote the selected movable-AP subset in relocation window w. In the Rabot scenario studied here, $|\mathcal{S}_w| = N$ and no always-on distributed APs are used, so the distributed comparison isolates the effect of changing the movable subset over time.

B. Relocation Heuristic

The movable distributed strategy is implemented as a windowed receding-horizon controller. Let $\{t_n\}_{n=0}^{N_t-1}$ denote the simulated snapshot times and let T_{rel} denote the relocation interval. The trajectory is partitioned into disjoint decision windows

$$\mathcal{W}_w = \left\{ n \in \{0, \ldots, N_t - 1\} : \left\lfloor \frac{t_n - t_0}{T_{\text{rel}}} \right\rfloor = w \right\}. \tag{6}$$

At the beginning of each window w, the controller selects a subset $\mathcal{S}_w \subseteq \mathcal{C}$ with cardinality $|\mathcal{S}_w| = N_{\text{mov}}$, where N_{mov} is the movable-AP budget. The fixed distributed mode uses one time-invariant subset, whereas the movable mode recomputes $\mathcal{S}_w$ at every relocation instant.

The online decision rule is intentionally local and does not optimize a full radio map. For a candidate site $c \in \mathcal{C}$ with horizontal position $\mathbf{x}_c \in \mathbb{R}^2$, define the local neighborhood $\mathcal{N}_K(c, w)$ as the set of the K nearest UE-snapshot pairs in the horizontal plane within window w,

$$\mathcal{N}_K(c, w) = \text{KNN}_K(\mathbf{x}_c; \{\mathbf{r}_{u,n} : u \in \mathcal{U}, n \in \mathcal{W}_w\}), \tag{7}$$

where $\mathbf{r}_{u,n}$ denotes the horizontal position of UE u at time index n. For a subset $\mathcal{S} \subseteq \mathcal{C}$, the support region used by the controller is the union

$$\mathcal{N}_K(\mathcal{S}, w) = \bigcup_{c \in \mathcal{S}} \mathcal{N}_K(c, w). \tag{8}$$

This restriction makes the relocation rule sensitive to the CSI observed near the candidate sites rather than to globally averaged scene statistics.

For every queried subset $\mathcal{S}$, the implementation computes the corresponding AP-to-UE CSI/SINR segment over the current window and stores it in a per-subset cache. If $\gamma_{u,n}(\mathcal{S})$ denotes the best-link SINR sample produced when subset $\mathcal{S}$ is active, the historical record available when deciding window w is

$$\mathcal{H}_w(\mathcal{S}) = \{(t_n, \gamma_{u,n}(\mathcal{S})) : u \in \mathcal{U}, n \in \mathcal{W}_j, j = 0, \ldots, w\}. \tag{9}$$

When the same subset is revisited later, previously computed window segments are reused and only the new current-window segment is appended. Thus, the historical objective remains consistent while avoiding repeated ray-tracing evaluations for past windows.

Historical CSI is discounted through exponential aging. If a sample was observed at time t_n and the relocation decision for window w is taken at t_w^{end}, the associated weight is

$$\omega_n^{(w)} = \exp\left[-\lambda\left(t_w^{\text{end}} - t_n\right)\right], \tag{10}$$

with λ the configured decay rate. The local control objective is then the weighted empirical 10th percentile of the SINR samples retained by the local neighborhoods across all observed windows,

$$J_w(\mathcal{S}) = Q_{0.1}^{\omega}\left(\left\{\gamma_{u,n}(\mathcal{S}) : (u,n) \in \bigcup_{j=0}^{w} \mathcal{N}_K(\mathcal{S}, j)\right\}\right), \tag{11}$$

where $Q_{0.1}^{\omega}(\cdot)$ denotes the weighted 10th-percentile operator induced by $\omega_n^{(w)}$. The use of a lower-tail percentile, rather than a mean, makes the controller deliberately conservative with respect to persistently weak local links.

The subset search itself is greedy. Starting from $\mathcal{S}_w^{(0)} = \varnothing$, the controller adds one candidate at a time according to

$$c_r^{\star} = \arg\max_{c \in \mathcal{C} \setminus \mathcal{S}_w^{(r-1)}} J_w\left(\mathcal{S}_w^{(r-1)} \cup \{c\}\right), \tag{12}$$

$$\mathcal{S}_w^{(r)} = \mathcal{S}_w^{(r-1)} \cup \{c_r^{\star}\}, \qquad r = 1, \ldots, N_{\mathrm{mov}}, \tag{13}$$

and finally sets $\mathcal{S}_w = \mathcal{S}_w^{(N_{\mathrm{mov}})}$. Consequently, the current implementation is a forward-selection heuristic rather than an exhaustive combinatorial optimizer.

After the subset has been chosen, the selected geometric candidate sites are converted into physical relocations of labeled movable APs. Let $\mathbf{p}_m^{(w-1)}$ denote the previous-window position of movable AP m and let $\mathbf{q}_j^{(w)}$ denote the coordinates of the candidates in the newly selected subset. The controller solves the linear assignment problem

$$\pi_w^{\star} = \arg\min_{\pi \in \Pi_{N_{\mathrm{mov}}}} \sum_{m=1}^{N_{\mathrm{mov}}} \left\| \mathbf{p}_m^{(w-1)} - \mathbf{q}_{\pi(m)}^{(w)} \right\|_2^2, \tag{14}$$

where $\Pi_{N_{\mathrm{mov}}}$ is the permutation set on $\{1, \ldots, N_{\mathrm{mov}}\}$. This minimum-displacement matching preserves AP identities across windows and yields schedule traces that correspond to physically meaningful AP trajectories instead of arbitrary candidate relabelings.

Once all windows have been simulated, the concatenated movable-strategy trajectory is evaluated using the same trajectory-level summary score as the static baselines,

$$F = -\alpha P_{\mathrm{out}} + \beta P_{10} + \eta T_{\mathrm{peer}}, \tag{15}$$

where P_{out} is the fraction of user snapshots below the configured SINR threshold, P_{10} is the trajectory-wide 10th percentile, and T_{peer} is a peer-weighted mean SINR term derived from UE-to-UE CSI. The coefficients (α, β, η) correspond to the configured outage, percentile, and peer tie-break weights. This final score is used for reporting and mode comparison; the online relocation controller itself is driven by $J_w(\mathcal{S})$, i.e., by local historical CSI only. The above formulation defines the relocation mechanism used throughout the Rabot simulations.

IV. Simulation Setup

The Rabot outdoor scenario is defined in `scenarios/rabot.yaml`. The scene is derived from an OpenStreetMap (OSM) boundary around the KU Leuven Gent Campus Rabot environment. Candidate distributed-AP positions are generated along building walls at low height, one rooftop proxy is reserved for the central reference strategy, and user mobility is simulated over the outdoor walkable area with mixed pedestrian and bicycle profiles.

TABLE I: Key Rabot scenario parameters used in the Rabot simulations.

Parameter	Value
Carrier frequency	3.5 GHz
Bandwidth	100 MHz
User height	1.5 m
Number of users	10
Simulation horizon	3600 s
Time step	5 s
Distributed APs per strategy	6
Fixed distributed APs	0
Candidate wall height	1.5 m
Candidate wall spacing	18 m
Minimum candidate spacing	12 m
Candidate wall offset	0.25 m
Relocation window	600 s
Historical CSI decay rate	0.011 55 s^{-1}
Local neighborhood size	10 nearest UE snapshots per candidate
Initialization seed	11
Compared strategies	central proxy, fixed distributed, movable distributed
Coverage maps	disabled in current Rabot config

TABLE II: Placement-strategy definitions used throughout the article.

Strategy	Description
`central_massive_mimo`	Single rooftop proxy on the building nearest the area center, oriented toward the center. Use as central reference only.
`distributed_fixed`	One distributed subset is selected from wall candidates and kept unchanged for the full run.
`distributed_movable`	The distributed subset is reselected every relocation window using local CSI with exponential decay over historical samples.

The simulation workflow used by the paper matches the repository workflow:

1) build the outdoor scene and walk graph from OSM,
2) generate wall-mounted candidate APs and the single rooftop proxy site,
3) simulate user trajectories over the walkable outdoor area,
4) compute path-based CSI and explicit SINR terms,
5) compare the central, fixed-distributed, and movable-distributed strategies,
6) export reusable result files,
7) generate analysis figures in a separate postprocessing step.

This section documents the reproducible processing chain from scene generation to postprocessed figure export.

The figure set in Sections V to VII is linked directly to generated artifacts. In the present workflow, scene-layout figures live in `outputs/rabot/`, while SINR, ESR, and schedule figures live in `outputs/rabot/postprocessing/`. This output contract ensures that the reported figures can be regenerated from the stored simulation products.

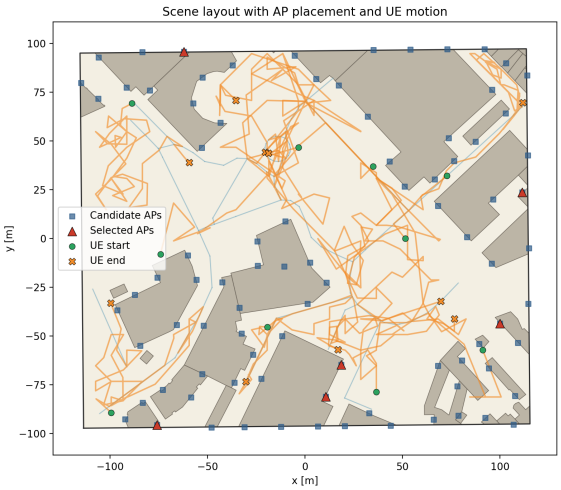
For the current Rabot instantiation, the relocation controller starts from a single seed constellation

drawn from the wall-mounted candidate pool using the configured random seed. That seed constellation is frozen for the `distributed_fixed` baseline and also provides the initial labeled movable-AP positions for `distributed_movable`. Thereafter, every relocation window applies the greedy local-CSI subset search from Section III and then solves a minimum-displacement assignment so that schedule traces, relocation distances, and move fractions correspond to the motion of persistent AP identities rather than to arbitrary reordering of candidate sites.

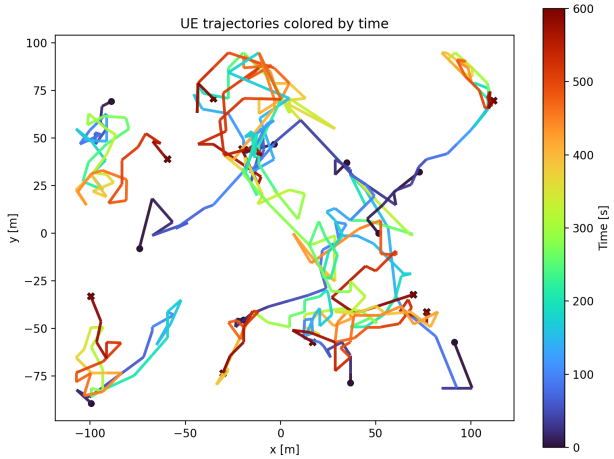
V. Scenario Geometry and Mobility Results

This result block summarizes the simulated environment before presenting radio-performance metrics. It reports the available candidate mounting locations, the selected deployment, and the spatial distribution of user trajectories over the Rabot area.

Figure 2 provides the geometric context for the subsequent SINR, ESR, and relocation analyses. The scene-layout panel reports the candidate-site pool and the selected constellation, whereas the trajectory colormap reports the spatial spread of the simulated user population.



(a) Top-down Rabot scene layout with candidate sites, selected sites, and simulated mobility overlays.



(b) Trajectory colormap used to visualize the temporal spread of user motion through the walkable space.

Fig. 2: Geometry results for the Rabot scenario. The panels show the selected deployment in the urban layout and the temporal distribution of user motion over the walkable area.

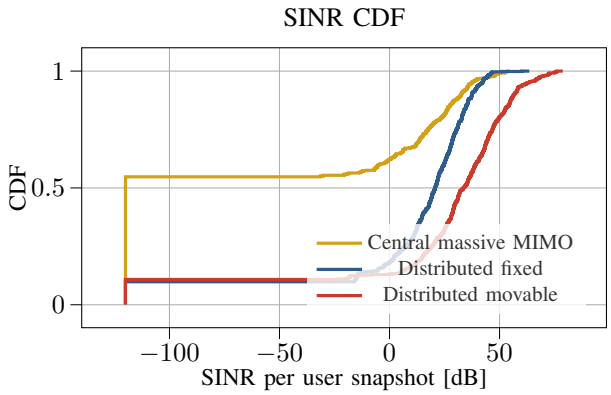
VI. SINR- and ESR-Oriented Results

This section reports the radio-performance figures generated by the postprocessing workflow. The panels summarize global SINR statistics, user-level averages, temporal ESR evolution, aggregate ESR distributions, and worst-user behavior.

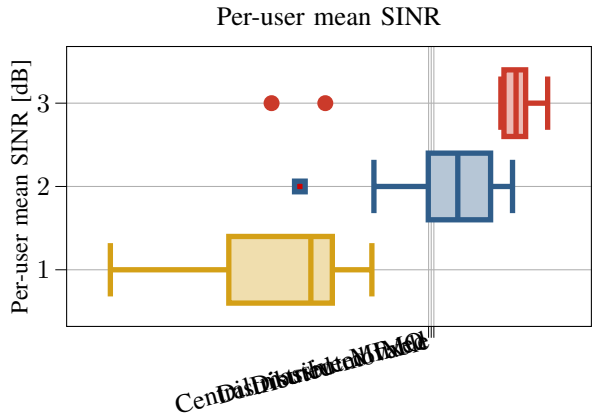
Together, these figures provide complementary views of trajectory-level performance, user-level variability, temporal evolution, and lower-tail robustness across the compared placement strategies.

TABLE III: Mapping between article figures and postprocessed artifacts.

Figure in article	Postprocessed artifact
Global SINR distribution	`postprocessing/sinr/sinr_cdf_publication.te`
Per-user mean boxplot	`postprocessing/sinr/per_user_mean_sinr_boxp`
ESR time series	`postprocessing/sinr/esr_timeseries.tex`
ESR CDF	`postprocessing/sinr/esr_cdf.tex`
Worst-user SINR view	`postprocessing/sinr/worst_user_sinr_timese`

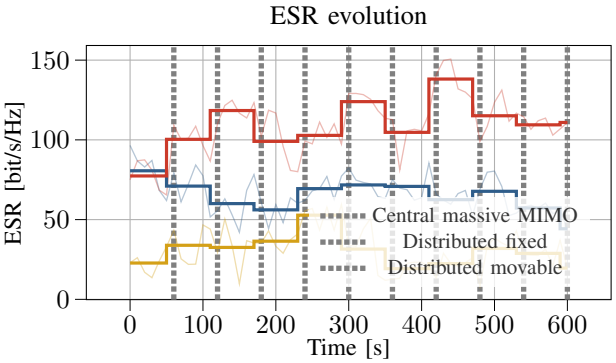


(a) Empirical SINR distribution across all user snapshots and compared placement strategies.

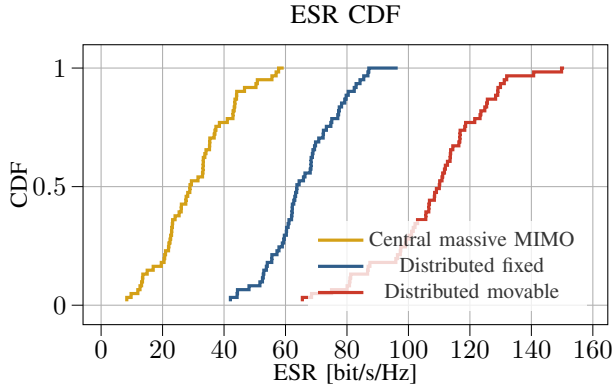


(b) Per-user mean SINR boxplot across strategies.

Fig. 3: First pair of radio-performance results. The left panel reports the global SINR distribution, and the right panel reports the distribution of per-user mean SINR.



(a) ESR over time, including relocation markers and window-level step behavior.



(b) ESR cumulative distribution function (CDF) across all evaluated snapshots.

Fig. 4: Second pair of radio-performance results. These panels report the temporal evolution of ESR and its global empirical distribution.

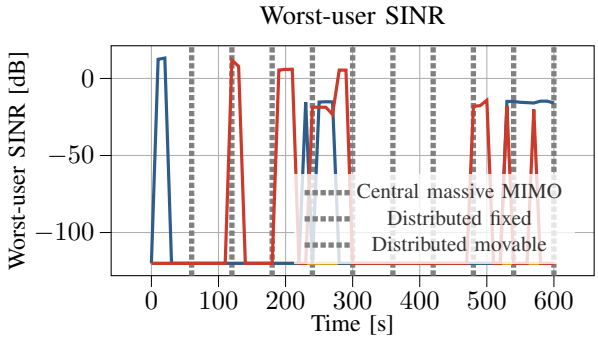
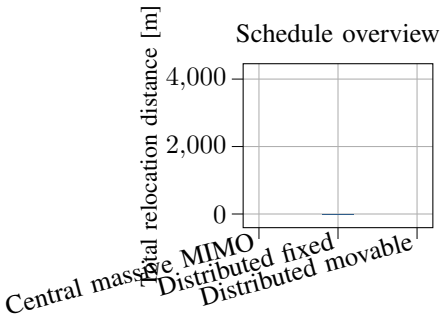


Fig. 5: Worst-user SINR over time.

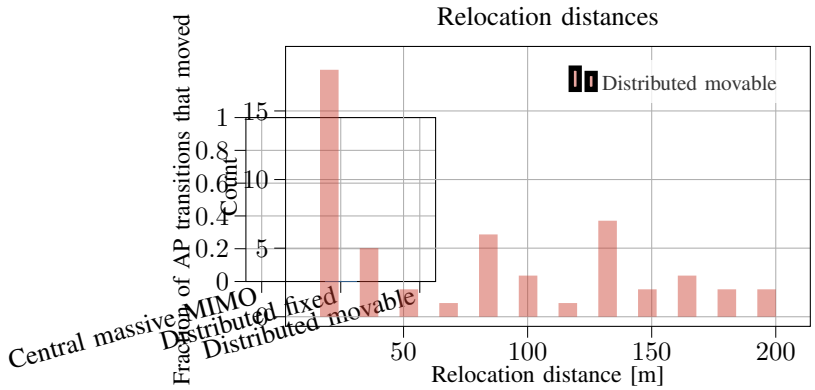
VII. Relocation and Practicality Results

The relocation figures complement the radio-performance results by quantifying the physical reconfiguration burden associated with movable infrastructure. The scheduling outputs report how often the movable APs change position and how much displacement the resulting schedules induce.

Together, the panels in Fig. 6 characterize the relocation activity associated with the evaluated placement strategies through aggregate movement summaries and relocation-distance distributions.



(a) Strategy-level overview of total relocation distance and move fraction.



(b) Distribution of relocation distances across observed AP moves.

Fig. 6: Relocation-cost figures used to interpret how much physical infrastructure movement is required by the compared placement strategies.

VIII. Discussion and Current Limits

The present study has several modeling and evaluation limits that define the scope of the reported results.

The main limitations are the following:

- Only a single outdoor scenario is considered.
- The `central_massive_mimo` comparison point is a rooftop proxy, not a full placement model for one true co-located massive-multiple-input multiple-output (MIMO) array.
- The current Rabot configuration disables coverage-map exports, so the paper centers on trajectory-level SINR and schedule metrics.
- Citizen participation is abstracted as movable candidate positions and scheduler actions rather than being validated experimentally.
- Sensitivity to random seeds, scenario perturbations, and broader deployment-cost models is not yet characterized.

These limitations motivate extensions toward multi-scenario studies, sensitivity analyses, coverage-map reporting, and eventual experimental validation in co-created deployments.

IX. Conclusion

This article presented a simulation-based study of low-height distributed-MIMO access networks with repositionable infrastructure in an outdoor urban scenario. The paper combined a system model, a history-aware relocation algorithm, a reproducible Rabot simulation workflow, and postprocessed geometry, SINR, ESR, and relocation figures derived from exported artifacts.

Within its current scope, the framework supports systematic comparison of a central rooftop proxy, a fixed distributed deployment, and a relocation-capable distributed deployment under a common channel model and mobility process. Natural next steps are multi-scenario studies, coverage-map analysis, and experimental validation in the broader Cocoon program.